

Institute of Actuaries of India

Subject CB1-Business Finance

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

IAI	CB1-1120
<u>Solution 1:</u> B	[2 Marks]
<u>Solution 2:</u> C	[2 Marks]
<u>Solution 3:</u> B	[2 Marks]
<u>Solution 4:</u> D	[2 Marks]
<u>Solution 5:</u> C	[2 Marks]
<u>Solution 6:</u> C	[2 Marks]
<u>Solution 7:</u> A	[2 Marks]
<u>Solution 8:</u> C	[2 Marks]
<u>Solution 9:</u> B	[2 Marks]
<u>Solution 10:</u> D	[2 Marks]

Solution 11:

- i) 0.4/5 Net 20 is a trade discount structure wherein 0.4% discount is offered by company B Ltd. if A Ltd. pays within 5 days, otherwise company A Ltd. has 20 days to settle its dues. [1]

ii) **Settlement options:**

A Ltd. has following options:

- Pay INR 96,600 on day 5 i.e. avail the discount on last possible day
- Pay INR 1,00,000 on day 20 i.e. do not avail the discount but defer the payment as much as possible
- Pay INR 1,00,000 on any other day between 5th and 20th, let's say day 15

Analysis Required:

To evaluate the most efficient repayment time for A Ltd, we need to calculate cost of trade credit for options b) and c)

The effective annual cost of this trade credit is calculated by formula:

$$\text{cost of trade credit} = \left(1 + \frac{\text{discount}}{1 - \text{discount}} \right)^{\left(\frac{365}{\text{payment day} - \text{discount period}} \right)} - 1 \text{ pa}$$

Option b) Discount = 0.4% | Payment Day = 20 | Discount Period = 5

Cost of trade credit (on day 20) = 10.2%

Option c) Discount = 0.4% | Payment Day = 15 | Discount Period = 5

Cost of trade credit (on day 15) = 15.8% [2.5]

Conclusion:

- Clearly, the cost of trade credit increases on early payment during the non-discount period. Hence, option c) or any settlement during 5th to 20th day shall be inefficient for A Ltd.

- Similarly, if A Ltd. intends to exercise the discount, it should make the payment on day 5 and not earlier.
- Choice of making payment on day 5 vs day 20 shall depend on potential of A Ltd. to utilize the funds. If it can generate returns higher than 10.2% p.a. during 20 day period, it should not avail the discount offered by B Ltd.

[1.5]

[5 Marks]**Solution 12:**

Revised premiums and variable costs in the flexed budget are calculated as:

$(120/100) \times \text{Original Budget Figures}$

For support staff salaries and rent, no change is required to the original budget as these are fixed in nature.

This results in a flexed budget of:

Component	Original Budget	Actual	Flexed Budget
Number of Policies	100	120	120
Premiums	1,00,000	1,00,000	1,20,000
Salaries	35,000	25,000	33,000
Support staff	21,000	15,000	21,000
Sales staff	14,000	10,000	12,000
Commission	30,000	35,000	42,000
Rent	20,000	22,000	20,000
Profit	15,000	18,000	25,000

The total adverse profit variance for the quarter, as compared to the flexed budget, is 7,000. This is made up of:

- An adverse sales revenue variance of 20,000. This may be a result of, for example, increased competition in the sector or the competitors cutting prices to boost sales.
- A favourable variance of 8,000 in the salaries of sales staff. This may be a result of, for example, reducing a component of the compensation of the sales staff, owing to lower volumes.
- An favourable variance of 7,000 in commissions as these are directly linked to sales. Further, this may be because products sold attract lower commissions than the planned products.
- An adverse variance of 2,000 in rental costs. This may be a result of, for example, an incorrect initial forecast, revision of contracts of few branches, opening of new offices, etc.

[5 Marks]**Solution 13:**

The current value of machinery in the books of X Ltd. shall be INR 50 lakhs*. Making accounts on fair-value basis shall result in a revaluation gain on machinery of INR 30 lakhs (=Market value less Book value).

In the statement of financial position, this will result in the following changes:

- The asset value will be replaced with the new revalued figure i.e. INR 80 lakhs

- Depreciation to date will be set to zero. Annual depreciation will now be based upon the revalued figure, with the asset written off over its estimated useful lifetime.
- X Ltd. may revise its expectations of the useful remaining lifetime of the asset and the residual value
- A revaluation reserve will be included in the equity section under the heading 'Other reserves'. This revaluation reserve will be the difference between the revalued asset and the book value of the asset at the date of revaluation (i.e. INR 30 lakhs)

In the statement of comprehensive income, a gain on revaluation shall be shown as one element of 'other comprehensive income'.

The revaluation will also be listed in the statement of changes in equity.

*This valuation is based on the following assumptions:

- X Ltd. used straight line method for depreciation
- There is no change in useful lifetime estimated at the time of purchase of the asset
- Residual value of the machinery at the end of 10th year is zero.

[5 Marks]

Solution 14:

Sustainability reporting enables organisations to measure, understand and communicate the economic, social and environmental effects of their activities.

A sustainability report also presents the organisation's goals, values and model of governance.

A sustainability report can be produced:

- as a non-financial report, or
- as an intrinsic element of integrated reporting

Advantages:

- It compels organisations to recognise that actions taken now have implications for the future
- It helps organisations to consider and communicate their sustainability vision and strategy in the context of their overall goals
- It recognises the variety of stakeholders that are involved in the organisation and encourages businesses to consider the overall public interest in the decision-making process
- It demands greater transparency, which then enables and empowers businesses, governments, consumers and citizens to make informed decisions
- It enhances the company's image and reputation
- It helps organizations to attract and retain employees
- It creates competition within the industry

Disadvantages

- There are practical difficulties of measurement and projection; e.g. estimating the effect of water pollution and projecting these effects into the future not an easy task
- There is a danger that companies report the good news and hide the bad. Such behaviour might lead to a lack of credibility in the reports

[5 Marks]

Solution 15:

The probability tree would involve the following steps:

mapping the possible choices, beginning from the initial project decision and branching out to represent all the possible subsequent options [1.25]

assigning estimated cashflows associated with each future possible choice estimating the probabilities associated with each future cashflow [1.25]

using standard expected value calculations, incorporating both the time value of money and the probabilities, to assess the optimal choices in each future time period based on the knowledge of the intervening events [1.25]

working backwards from the latest decision point to the present day in order to establish the best (e.g. highest NPV) route to follow at the outset [1.25]

[5 Marks]

Solution 16:

Stakeholders have conflicting interests due to the commercial realities of the business. [1]

For example, corporate social responsibility is good for the public but may be costly to the shareholders. Therefore, benefits accruing to one stakeholder group may impose costs on another. This may be particularly acute when the company has direct contact with both parties, as when improving employees' benefits may be expensive for the shareholders. [1.5]

A related issue arises from the fact that the interests of different providers of finance may be in direct conflict. [1]

For example, the shareholders enjoy the upside from risky investments but the lenders do not. A risky investment could be good for the shareholders, but might undermine the security of the lenders. Similarly, taking on additional debt in order to finance growth may be in the shareholders' interests, but that could dilute the security enjoyed by lenders. [1.5]

[5 Marks]

Solution 17:

One interpretation of the Modigliani and Miller model is that shareholders may suffer a loss because of tax if a company changes its dividend policy.

ABC's shareholders clearly prefer capital gains to dividends and so they will suffer a significant tax charge if a dividend is paid without giving them the opportunity to make arrangements.

ABC should not retain significant cash balances

if there are no investment opportunities available in order to put them to work [1] It may be better for ABC to inform the markets that the industry is not in a suitable condition to make additional investments and so it will be making a significant dividend payment in due course

The shareholders can then realise some or even all of their equity by selling the shares in advance of the dividend, selling to shareholders whose circumstances lead to them having a preference for dividend income.

[5 Marks]

Solution 18:

1. acting as the “lender of last resort”
2. acting as the bankers’ bank (this allows easy daily settlement of inter-bank debt)
3. Maintaining foreign reserves, and using them to influence exchange rates.
4. Setting of monetary policy
5. Setting up banking regulations
6. Implementation of government borrowing
7. Printing and minting of notes and coins

Impact on personal sector:

- A major direct effect of reduced interest rates for the personal sector is likely to arise via the decrease in mortgage loan interest payments, which will increase disposable income and hence higher personal sector expenditure.
- Consumers’ expenditure may also be encouraged by lower rates on credit facilities and lower rates of interest may also encourage lower levels of saving.

Impact on business sector

- The impact on the business sector is likely to be positive since capital investment and economic growth prospects will be high.. This is due to the reduced opportunity cost of committing funds for investment and the lower cost of borrowing.
- With lower interest rates, many projects being considered will demonstrate a positive net present value. Investment is therefore likely to be higher

[5 Marks]

Solution 19:

i) Accounting Concepts

a) Cost Concept and Going Concern Concept:

Cost concept implies that non-current assets shall appear in the statement of financial position at their original cost less depreciation to date, subject to a possible impairment write-down.

According to Going concern concept, financial statements are prepared assuming that a business will continue indefinitely in its present form. [2]

b) Relationship between the two accounting concepts:

The going concern concept, where it is assumed that a business will continue indefinitely in its present form, acts as a justification for the limitations imposed by the cost concept because there is little harm in reporting historical figures for value if the assets concerned are unlikely to be sold in the immediate future. [1]

c) **Ease in preparation of accounts:**

This simplifies the preparation of financial statements because:

- There is no need to make estimates of the market value of all the company's assets and liabilities
- Not claiming to estimate market values also leaves the preparers and auditors of the accounts less open to challenge in the event that their estimates prove to be incorrect
- Non-current assets that are purchased can be depreciated on a straight-line basis over the useful life of the asset, because we assume that the asset will continue to be used for its useful life
- Potential errors in short-term estimates can be tolerated because the figures will resolve themselves over time
- Inventory, which often has zero value on wind up, can be included in the financial statements at original cost until it is used and generates revenue [3]

ii) **Missing Values | Accounting Ratios**

Current Ratio = Current Assets/ Current Liabilities

$$M1 = 128/88 = 1.45$$

Quick Ratio = (Current Assets – Inventories) / Liabilities

$$M2 = (145-49)/112 = 0.86$$

Inventory Turnaround = (Inventory/Cost of Sales) * 365 days

$$M3 = 55/532 * 365 = 38 \text{ days}$$

Trade Payable Turnaround = (Trade Payable/Credit Purchases) * 365 days

$$M3 = 75/510 * 365 = 54 \text{ days}$$

(Assuming all the purchases have been made on credit)

Revised Table on Financials:

<u>Financial Ratio/FY</u>	2017	2018	2019
Current Ratio	1.45	1.29	1.19
Quick Ratio	0.98	0.86	0.82
Inventory Turnover	33	34	38
Trade Receivable Turnover	28	32	37
Trade Payable Turnover	54	63	81

[2]

iii) **Analysing Trends in Liquidity Ratios**

- If a company faces liquidity problems, it could result in failure of the company because of its inability to pay debts.
- A static position with regards to liquidity, however, can prove misleading when viewed in isolation.
- Analysis of trends can reveal the effectiveness of strategies for managing liquidity. For example:
 - Low but stable ratio for many years may mean that the poor ratio may be adequate for survival of that company/industry.
 - High liquidity ratios may mean resources are not being utilized optimally.

- Hence, reducing liquidity ratio may suggest a cash-surplus company's strategy to aim reduction in liquidity in order to put those resources to good use

[3]

iv) **Liquidity and Efficiency of ABC Ltd.**

a) **Liquidity:**

- Following ratios indicate the liquidity position of a company:
 - **Current Ratio** compares money due to be received soon with money due to be paid soon
 - **Quick ratio** works similar to current ratio; however, it uses only the cash or near-cash items in the statement of financial position (as stocks may take a while to sell)
- [1]
- A current ratio of ~1.2 in 2019 indicates that ABC is able to cover its short-term debt. However, the ratio appears low (against an informal benchmark of 2)
 - A quick ratio of ~0.8 indicates that the company may struggle to pay its creditors
 - ABC's liquidity is a matter of some concern as the ratios are not only low but also declining
 - The decreasing current and quick ratios may suggest that the company is running into increasing difficulties and that it may not be able to sustain a quick ratio of less than 1.0
 - The cash balance has fallen from 20,000 in 2017 to 11,000 in 2018 and further down to 2,000 in 2019.
 - This suggest that some cashflow problems have arisen with the increased revenues.
 - The key issue is that Fratton is steadily running out of cash and will soon cross over to an overdraft situation
- [2.5]

Efficiency:

- Efficiency ratios give an insight into how effectively the company is managing the components of working capital (current assets less current liabilities) [0.5]
- Inventory Turnover Period assesses how much inventory the company holds in relation to the scale of the company's operations.
 - In other words, it measures how quickly a company is selling its output. [0.5]
 - The increasing number of days in inventory turnover suggest that more and more cash is being tied up in these assets, despite the fact that they do not yield any return. [0.5]
- Trade Receivable Turnover indicates the average number of days' credit that is extended to customers paying by credit [0.5]
 - The increasing number of days in receivable turnover suggests that company is not operating its credit control function effectively. [0.5]
- Trade Payable Turnover indicates the average number of days' credit that a company has from its suppliers. [0.5]
 - The trade payables turnover period is also increasing over the years. This means that ABC Ltd. may have been able to negotiate good credit terms from its suppliers. [0.5]
- Worsening of inventory and receivable turnover ratios over the years suggest that as the size of the business has grown, the existing systems and processes are not working so well.
- Despite increasing inventory and receivable turnover ratios over the years, the company has effectively reduced its working capital cycle.
- This is because trade payable turnover ratio is increasing at a faster rate as compared to the increase in inventory and receivable turnover ratios.
- This also implies that credit from suppliers is being used as a source of short-term finance, thereby reducing the impact of poor liquidity position.

- The danger is that ABC Ltd. might take excessive advantage of suppliers, which could result in strained relations in a longer run.
- Hence, this strategy to tie up cash in inventories and using credit as source of finance might not be sustainable but appears to be efficient in short term.

[2]

[20 Marks]**Solution 20:****i)**

Market Capitalization	400	Crore
Debenture		
Face value	250	
Market value	220	
Coupon Rate	5%	
Redeemable	5	years

Tax Rate	30%
Risk Free Rate (Rf)	4%
Ungeared Beta	1.4

Market Rate (Rm)	12%
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Geared beta= ungeared beta × (1 + debt: equity ratio × (1 –tax rate))

Geared Beta	1.939	[2]
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Cost of Equity= ke = Rf+ (geared beta × Rm- Rf)

Cost of Equity	19.51%	[2]
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ii)

Coupon payment on debt =	12.5	
NPV of debt at 6%	19.4	=-220+(12.5*4.212)+(250*0.747)

NPV of debt at 10%	-17.3625	=-220+(12.5*3.791)+(250*0.621)
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Yield is approximately half way between 6% and 10% = 8% (to nearest whole %)

Kd=		
5.6000%	=8%*(1-tax rate)	[4]

iii)

$$=(220/(220+400)*Kd)+400/(220+400)*Ke$$

$$WACC= 0.1457$$

[2]

iv) Rights Issue

A rights issue is where a company offers further shares, at a given price, to existing shareholders in proportion to their existing holdings.

The price will be at a discount to the current share price.

The main effects of a successful rights issue are:

- New shares are created
- New money is raised for the company
- The total value of the whole company should be increased by the extra money raised
- The price per share will fall depending on the extent of the discount and the number of new shares issued.

[3]

Convertible Unsecured Loan Stock

Convertible unsecured loan stocks are unsecured loan stocks which give the right to convert into ordinary shares of the company at a later date.

The characteristics of a convertible security in the period prior to conversion are a cross between those of fixed-interest stock and ordinary shares. As the likely date of conversion (or not) gets nearer, it becomes clearer whether the convertible will stay as loan stock or become ordinary shares. As this happens, its behaviour becomes closer to that of the security into which it will convert.

The convertible loan stock will have a stated annual interest payment (paid in twice-yearly instalments).

[3]

Further Factors that Company should consider

- The cost of raising the finance
The company should consider the initial issue cost, plus the return that will have to be paid to the investors.
- The tax position
Interest on debt finance is tax deductible, whereas dividend payments are not.
- The risk to the company
Interest on debt has to be paid, whereas dividends are paid at the discretion of the directors.
- The effect on control
Debt holders have no vote, but shareholders do.
- Flexibility of finance
Share capital is usually irredeemable (though share buybacks are possible) whereas debt is redeemable and can be issued at various terms.

[4]

[20 Marks]
